

$$a_n = \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}} \quad n \geq 1$$

$$\lim_{n \rightarrow \infty} a_n = ?$$

$$a_1 = \sqrt{2}$$

$$a_2 = \sqrt{2 + \sqrt{2}}$$

$$x = \sqrt{2 + \sqrt{2 + \dots}}$$

$$x^2 = 2 + \sqrt{2 + \dots}$$

$$x^2 = 2 + x$$

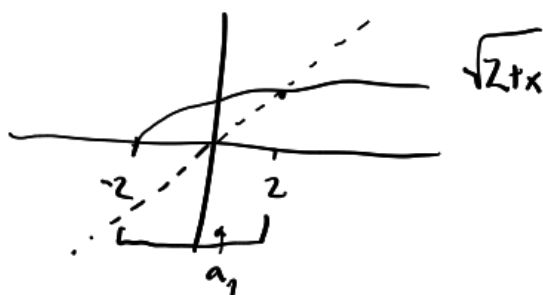
$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$\boxed{x_1 = 2}$$

$$x_2 = -1 \text{ nerealiz.}$$

$$a_{n+1} = \sqrt{2 + a_n}$$



$$\sqrt{2 + a_n} \leq \sqrt{2 + a_{n+1}}$$

$$2 + a_n \leq 2 + a_{n+1}$$

$$a_n \leq a_{n+1}$$

IP.

$$a_n \leq 2 \rightarrow a_{n+1} \leq 2$$

$$\sqrt{2 + a_n} \leq 2$$

$$2 + a_n \leq 4$$

$$\underline{a_n \leq 2}$$

6. $a_0 = 0$
 $a_1 = 1$

$$a_{n+2} = a_{n+1} + a_n \quad n \in \mathbb{N}$$

Dołaci spl. člen

$$a_n = \lambda^n$$

$$\lambda^{n+2} = \lambda^{n+1} + \lambda^n$$

$$\lambda^2 - \lambda - 1 = 0$$

$$\lambda_{1,2} = \frac{1 \pm \sqrt{5}}{2}$$

$$a_n = A \left(\frac{1 + \sqrt{5}}{2} \right)^n + B \left(\frac{1 - \sqrt{5}}{2} \right)^n$$

$$a_1 = 1$$

$$a_0 = 0$$

$$A \left(\frac{1 + \sqrt{5}}{2} \right) + B \left(\frac{1 - \sqrt{5}}{2} \right) = 1$$

$$A + B = 0$$

$$A = -B$$

$$A \left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} \right) = 1$$

$$A(\sqrt{5}) = 1$$

$$A = \frac{\sqrt{5}}{5}$$

$$B = -\frac{\sqrt{5}}{5}$$

$$a_n = \frac{\sqrt{5}}{5} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{\sqrt{5}}{5} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

$$\text{--- } \hat{C}e \quad \lambda_1 \neq \lambda_2 : a_n = A\lambda_1^n + B\lambda_2^n$$

$$\text{--- } \lambda_1 = \lambda_2 : a_n = (A + Bn)\lambda^n$$

Vorste

① ja

$$(4) \sum_{n=1}^{\infty} \frac{(2n)^n (2n)!}{(3n)!}$$

a)

$$\lim_{n \rightarrow \infty} \frac{(2n+2)^{n+1} (2n+2)! (3n)! (n+1)(n+2)}{(3n+3)! (2n)^n (2n)! (3n+1)(3n+2)(3n+3)} = \lim_{n \rightarrow \infty} \frac{2^{n+2} (n+1)^{n+2} (2n+1)}{2^n n^n (3n+1)(3n+2)3(n+1)}$$

$$= \frac{4}{3} \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n \lim_{n \rightarrow \infty} \frac{(2n+1)(n+1)}{(3n+1)(3n+2)} = \frac{8}{27} e < 1$$

⇓

po kvac. krit. konvergieren

$$b) \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} \cdot 2^{1-n}$$

$$\star n!! = n(n-2)(n-4) \dots$$

$$\frac{a_{n+1}}{a_n} = \frac{(2n+1)!! (2n)!! \cdot 2^{1-n-1}}{(2n+2)!! (2n-1)!! 2^{1-n}} = \frac{2n+1}{2n+2} 2^{-n-1+n} = \frac{2n+1}{2n+2} \cdot \frac{1}{2}$$

$$\lim_{n \rightarrow \infty} \frac{2n+1}{2n+2} \cdot \frac{1}{2} = \frac{1}{2} \Rightarrow \text{po kvoc. krit. konv.}$$

Primerjalni kriterij:

$$\sum a_n, \sum b_n$$

$$\text{če } \sum b_n \text{ konv. in } a_n \leq b_n \\ \Rightarrow \sum a_n \text{ konv.}$$

$$\text{če } \sum b_n \text{ diverg. in } b_n \leq a_n \\ \Rightarrow \sum a_n \text{ diverg.}$$

generating function

$$q^n \text{ konv. če } |q| < 1 \\ \text{div. če } |q| > 1$$

$$\frac{1}{n^\alpha} \text{ konv. če } \alpha > 1 \\ \text{div. če } \alpha \leq 1$$

3. Prim. krit., ali konv.?

$$a) \sum_{k=1}^{\infty} \frac{1}{k\sqrt{k+1}}$$

$$a_n = \frac{1}{n\sqrt{n+1}}$$

$$b_n = \frac{1}{n+\frac{1}{2}} = \frac{1}{n^{\frac{3}{2}}} \Rightarrow \sum_{k=1}^{\infty} \frac{1}{k\sqrt{k+1}} \text{ konv.}$$

$$b) \sum_{n=1}^{\infty} \frac{4n^2 + 3n + 1}{n^3 + 2n}$$

$$a_n = \frac{4n^2 + 3n + 1}{n^3 + 2n} = \frac{4n^2}{n^3 + 2n} + \frac{3n}{n^3 + 2n} + \frac{1}{n^3 + 2n} \leq \frac{4}{n^2} + \frac{3}{n^3} + \frac{1}{n^3}$$

$\Rightarrow \sum_{n=1}^{\infty} a_n$ konv., ker je vsota 3 konv. vrst konvergentna

$$c) \sum_{n=1}^{\infty} 2^n \sin\left(\frac{1}{3^n}\right)$$

$$a_n = 2^n \sin\left(\frac{1}{3^n}\right) \leq 2^n \frac{1}{3^n} = \left(\frac{2}{3}\right)^n$$

$\Rightarrow$ ker $\frac{2}{3} < 1$, $\sum a_n$ konvergira

$$\sum_{n=1}^{\infty} b_n = \frac{2}{3} \cdot \left(\frac{2}{3}\right)^{n-1} = \frac{2}{3(1-\frac{2}{3})} = 2$$

$$2. \sum_{n=1}^{\infty} \frac{1}{n^2 + n} = ?$$

$$\frac{1}{n^2 + n} = \frac{1}{n(n+1)} = \frac{A}{n} + \frac{B}{n+1} = \frac{1}{n} - \frac{1}{n+1}$$

$$A(n+1) + Bn = 1$$

$$An + A + Bn = 1$$

$$A = -B$$

$$A = 1$$

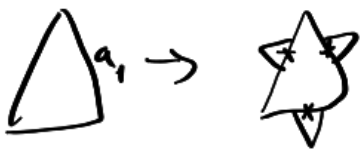
$$B = -1$$

$$s_n = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{n+1} = 1 - \frac{1}{n+1}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + n} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right) = 1$$

5. 0,5 Kochove sn.

0:



$$Q_1 = 3a_1 \underbrace{a_2}_{a_2}$$

$$Q_2 = 3 \cdot \left(\frac{4}{3}a_1\right)$$

$$Q_3 = 3 \cdot \underbrace{\frac{4}{3}a_2}_{a_3} = 3 \cdot a_3$$

$$Q_n = 3a_n = 3\left(\frac{4}{3}a_{n-1}\right) = \frac{4}{3}Q_{n-1} = \left(\frac{4}{3}\right)^{n-1} \cdot a_1 \cdot 3$$

